

SEEKING SUSTAINABLE AND INTEGRATED SOLUTIONS TO SOLVE CHALLENGES ACROSS THE AGRICULTURE AND FORESTRY VALUE CHAINS

Details

Background

Shell is committed to contributing to a net-zero world and has set a target to become a net-zero emissions energy business by 2050. Many of our customers have also announced their targets to achieve net-zero emissions. We will work together with our customers to change and grow demand for innovative low-carbon energy products and services.

Shell Agriculture & Forestry (A&F) aspires to enhance quality of life by supporting the production of food, clean air, water and soil by offering purposeful sustainable solutions to the A&F sector, creating value for all stakeholders.

Requirements

In this challenge, Shell A&F seeks partnership with solution providers and technology owners that offers novel and scalable solutions to decarbonise and offset greenhouse gas emissions (GHG). Specifically, Shell A&F aims to drive and provide solutions that monitor, reduce, capture and offset GHG in agricultural and forestry activities, through

Monitoring

- Solutions that enable **measurement and monitoring of surface and sub-surface soil carbon**, which includes soil organic carbon (SOC), soil inorganic carbon, total carbon, as well as a feature to diagnose and track soil carbon changes due to various dynamics such as seasonal fluctuations, agricultural practices, use of fertilisers and stimulants, etc.

Reduction and Capturing

- Solutions or means to **sequestrate or fix atmospheric carbon into agricultural/forest soil to improve soil health**. IoT-assisted regenerative practices, erosion control, restoration/remediation practices, use of microbial (fungi/bacteria) processes to reduce SOC losses and increase carbon input into soil, utilisation of novel energy crops or woody biomass from agricultural waste as feedstocks for energy production or conversion into valuable materials are among the solutions that may be considered.

Offsetting

- Solutions with a **digital platform to track, analyse and aggregate the GHG reduction, capture or offset activities** in a holistic manner, so as to facilitate the potential setup of a carbon credit marketplace, in which farmers from different regions may participate and generate additional incomes by selling carbon credits generated from emission reduction practices.
- Solutions that **improve the yield of food crops in urban agriculture**, such as AI/IoT assisted-indoor/outdoor vertical farming systems and technology that reduces energy, water, fertiliser and pesticides consumption, as well as through various controlled-environment agriculture (CEA) techniques.
- Solutions that **relate to green fertiliser manufacturing process technology** that is resource and energy efficient, as well as novel fertiliser technology that may help to improve agricultural soil health and boost crop productivity by providing/replenishing essential nutrients, including the macronutrients (nitrogen, phosphorus and potassium-based) and micronutrients. Greener production process, the use of zero-carbon/green ammonia as precursors and fertilisers that enable atmospheric carbon sequestration into soil are among the solutions that may be considered.

Solution providers and technology owners may also propose solutions or alternative approaches that align with the above interest areas. The solution provider must be able to conduct a pilot test at a suitable location determined by Shell A&F. Solutions with at least technology readiness level of TRL 4 (prototype in lab) is preferable. Shortlisted proposals will have to go through a selection process where solution providers will pitch their solution to Shell A&F and relevant business units for further evaluation.

- No specific on the cost target of the solution.

- Depending on the type and scalability of the solution and other assessment criteria, Shell A&F may consider a potential partnership to deploy the solution across the agriculture and/or forestry value chains where relevant.

Development Timeframe

- The solution provider must be able to conduct and complete the pilot test within a period of 6-12 months or within a suitable period determined by Shell A&F.

may provide necessary access and arrangement to a pilot test site for demonstration and validation of the solution.

Challenge
TechInnovation Challenge

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